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Universität Basel
Wirtschaftswissenschaftliche Fakultät

Machine Learning for Economists and Business Analysts

Lecturer: Prof. Anthony Strittmatter, Ph.D.

This PhD-level course is jointly organized by the University of Basel (WWZ) and UniDistance Suisse. It offers an intensive, hands-on introduction to modern machine learning methods that are increasingly central to empirical research in economics, business analytics and related fields.

Course format and locations

The course is taught in two parts:

- Part 1: March 11–13, 2026, Basel (WWZ, University of Basel)
- Part 2: March 25–27, 2026, Valais (UniDistance Suisse)

During their stay in Valais, participants will stay at Hotel Belalp in mixed-gender shared accommodation (dormitory-style rooms, up to 15 beds), with panoramic views of the Aletsch Glacier. Accommodation costs are covered. The setting supports focused learning as well as informal academic exchange in a unique environment.

Target audience

The course is open to doctoral students from:

- University of Basel
- UniDistance Suisse
- Swiss Network of PhD Courses in Economics and Finance
- EUCOR institutions

Course objectives and content

The course introduces key concepts and methods of machine learning for prediction, causal analysis, and data exploration, with an emphasis on when and how these tools are useful in economics and business contexts. The course is application-oriented and uses R throughout.

Participants will learn to:

- **Build predictive models (supervised learning):**
  Estimate and evaluate data-driven models, for example, predicting sales from price, quality, and market characteristics, with focus on out-of-sample performance.
- **Explore and structure complex data (unsupervised learning):**
  Identify hidden data patterns such as clusters of similar firms or consumers, and use dimensionality-reduction techniques to summarize and visualize high-dimensional datasets.
- **Estimate treatment effects (causal machine learning):**
  Study how interventions and policies affect outcomes (e.g., education on wages), including heterogeneous effects across subgroups, using flexible, data-adaptive methods.
- **Analyze sequential decision problems (reinforcement learning):**
  Understand learning and optimal decision-making in dynamic environments with feedback, relevant to areas such as pricing, targeting, and adaptive policy design.

Syllabus

- **Day 1 (Basel):** Fundamentals of Statistical Learning
- **Day 2 (Basel):** Regularized Regression
- **Day 3 (Basel):** Tree-Based Methods and Deep Learning
- **Day 4 (Valais):** Unsupervised Learning
- **Day 5 (Valais):** Causal Machine Learning, Student Presentations
- **Day 6 (Valais):** Reinforcement Learning

Student Presentations

During the second part of the course, students will present a project developed over the duration of the course.

Registration (deadline January 31, 2026)

- **University of Basel students:** Please enroll via email to GSBE (gsbe@unibas.ch).
- **UniDistance students:** Please register via email to Student Services (studentservices@unidistance.ch).
- **EUCOR and other Swiss mobility students (incl. FHNW):**
  You must first register as a mobility student at the University of Basel before the course starts (processing may take up to one week). After receiving your Basel login credentials, you can enroll via via email to GSBE (gsbe@unibas.ch).

The number of participants is limited. Preference will be given to PhD students from the University of Basel and UniDistance.

Contact

For questions about content or organization, please contact:
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Further details: https://vorlesungsverzeichnis.unibas.ch/de/vorlesungsverzeichnis?id=300268